

METACHRONOUS OVARIAN MALIGNANCY IN TRIPLE-NEGATIVE BREAST CANCER SUSPECTED OF HEREDITARY BREAST OVARIAN CANCER SYNDROME

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BACKGROUND

Triple-negative breast cancer (TNBC) is an aggressive breast cancer subtype strongly associated with hereditary cancer predisposition, particularly Hereditary Breast and Ovarian Cancer (HBOC) syndrome caused by germline BRCA1/BRCA2 pathogenic variants. Young-onset TNBC is a recognized indication for hereditary cancer risk assessment, and subsequent ovarian malignancy further increases suspicion of HBOC.

CASE PRESENTATION

A 32-year-old woman with an unavailable family history of malignancy presented with a palpable mass in the upper medial quadrant of the right breast. Histopathological examination confirmed TNBC (invasive carcinoma of no special type, grade II) with lymphovascular space invasion. Immunohistochemistry demonstrated estrogen receptor-negative, progesterone receptor-negative, HER2-negative status, and a Ki-67 proliferation index >80%. Following modified radical mastectomy, she received adjuvant chemotherapy and locoregional radiotherapy. One month later, contrast-enhanced CT demonstrated multiple osteolytic bone metastasis, and palliative radiotherapy with bone-targeted therapy was initiated.

Approximately one year later, markedly elevated CEA and CA15-3 levels, pleural effusion, and a pelvic mass prompted contrast-enhanced pelvic MRI, which demonstrated bilateral predominantly solid ovarian masses with heterogeneous enhancement and multiple bone metastasis. Based on the patient's history of metastatic TNBC and the imaging findings, the lesions were initially interpreted as bilateral ovarian metastasis. However, histopathological examination following total abdominal hysterectomy, bilateral salpingo-oophorectomy, and omentectomy revealed a mixed adult-type granulosa cell tumor with a moderately differentiated mucinous carcinoma component involving both ovaries, confirming a metachronous second primary ovarian malignancy.

DISCUSSION

This case highlights that atypical ovarian histology should not preclude hereditary cancer risk assessment in patients with young onset TNBC. Early recognition of hereditary cancer risk is essential to guide genetic counselling and germline testing, personalized management, optimize surveillance, and facilitate cancer risk assessment for at-risk family members.

CONCLUSION

This case highlights that atypical ovarian histology should not preclude hereditary cancer risk assessment in patients with young onset TNBC. Early recognition of hereditary cancer risk is essential to guide genetic counselling and germline testing, personalized management, optimize surveillance, and facilitate cancer risk assessment for at-risk family members.

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IMAGING FINDINGS

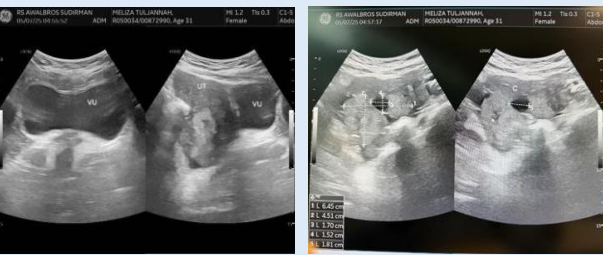


Figure 1. Abdominal ultrasound revealed a predominantly solid adnexal mass, irregular margin and a amorphous appearance, measuring 6,45 x 4,51 x 3,13 cm.

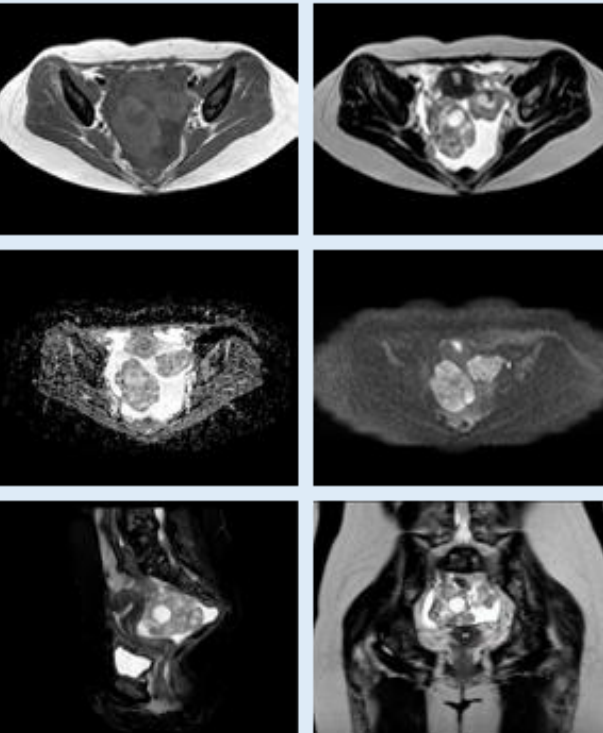


Figure 2. Pelvic MRI revealed inhomogeneous solid masses in both ovaries, measuring 6 x 5 x 4.5 cm (right) and 5.5 x 4 x 3.8 cm (left), with post-contrast enhancement. Accompanied by bone metastasis in the lumbar vertebral bodies, pelvis, and bilateral femoral head.

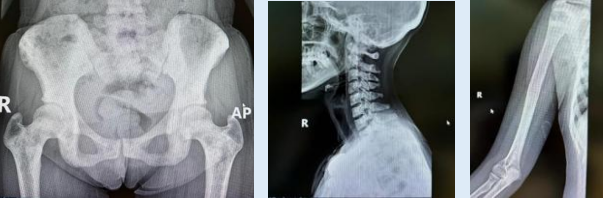


Figure 3. Bone survey revealed multiple lytic lesions involving the vertebral bodies of the cervical, thoracic, and lumbar spine, pelvis and right humerus



Figure 4. Whole spine CT revealed multiple lytic and sclerotic lesions involving the vertebral bodies of the cervical, thoracic, and lumbar spine, manubrium and body of the sternum

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